

10-(MAT/10_HL_Winter_2018_Q4)-Matrices

Consider the following system of equations where $a \in \mathbb{R}$.

$$\begin{aligned}2x + 4y - z &= 10 \\ x + 2y + az &= 5 \\ 5x + 12y &= 2a.\end{aligned}$$

- (a) Find the value of a for which the system of equations does not have a unique solution. [2]
- (b) Find the solution of the system of equations when $a = 2$. [5]



[Maximum mark: 5]

Find the values of x for which the vectors $\begin{pmatrix} 1 \\ 2\cos x \\ 0 \end{pmatrix}$ and $\begin{pmatrix} -1 \\ 2\sin x \\ 1 \end{pmatrix}$ are perpendicular,
 $0 \leq x \leq \frac{\pi}{2}$.

[illegible]

2-(MAT/10_HL_Winter_2012_Q9)-Vectors-Lines-Planes

Two boats, A and B , move so that at time t hours, their position vectors, in kilometres, are $\mathbf{r}_A = (9t)\mathbf{i} + (3 - 6t)\mathbf{j}$ and $\mathbf{r}_B = (7 - 4t)\mathbf{i} + (7t - 6)\mathbf{j}$.

- (a) Find the coordinates of the common point of the paths of the two boats. [4 marks]

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(b) Show that the boats do not collide.

[2 marks]

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3-(MAT/11_HL_Summer_2013_Q2)-Vectors-Lines-Planes

Consider the points $A(1, 2, 3)$, $B(1, 0, 5)$ and $C(2, -1, 4)$.

(a) Find $\vec{AB} \times \vec{AC}$. [4 marks]

(b) Hence find the area of the triangle ABC. [2 marks]

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4-(MAT/12_HL_Summer_2013_Q11)-Vectors-Lines-Planes

The vertices of a triangle ABC have coordinates given by $A(-1, 2, 3)$, $B(4, 1, 1)$ and $C(3, -2, 2)$.

- (a) (i) Find the lengths of the sides of the triangle.
- (ii) Find $\cos \hat{BAC}$. [6 marks]
- (b) (i) Show that $\vec{BC} \times \vec{CA} = -7\mathbf{i} - 3\mathbf{j} - 16\mathbf{k}$.
- (ii) Hence, show that the area of the triangle ABC is $\frac{1}{2}\sqrt{314}$. [5 marks]
- (c) Find the Cartesian equation of the plane containing the triangle ABC. [3 marks]
- (d) Find a vector equation of (AB). [2 marks]

The point D on (AB) is such that $\vec{OD}$ is perpendicular to $\vec{BC}$ where O is the origin.

- (e) (i) Find the coordinates of D.
- (ii) Show that D does not lie between A and B. [5 marks]



5-(MAT/10_HL_Winter_2013_Q11)-Vectors-Lines-Planes

Consider the points $A(1, 0, 0)$, $B(2, 2, 2)$ and $C(0, 2, 1)$.

- (a) Find the vector $\vec{CA} \times \vec{CB}$. [4]
- (b) Find an exact value for the area of the triangle ABC. [3]
- (c) Show that the Cartesian equation of the plane Π_1 , containing the triangle ABC, is $2x + 3y - 4z = 2$. [3]

A second plane Π_2 is defined by the Cartesian equation $\Pi_2: 4x - y - z = 4$. L_1 is the line of intersection of the planes Π_1 and Π_2 .

- (d) Find a vector equation for L_1 . [5]

A third plane Π_3 is defined by the Cartesian equation $16x + \alpha y - 3z = \beta$.

- (e) Find the value of α if all three planes contain L_1 . [3]
- (f) Find conditions on α and β if the plane Π_3 does **not** intersect with L_1 . [2]



6-(MAT/12_HL_Summer_2014_Q6)-Vectors-Lines-Planes

PQRS is a rhombus. Given that $\vec{PQ} = \mathbf{a}$ and $\vec{QR} = \mathbf{b}$,

- (a) express the vectors $\vec{PR}$ and $\vec{QS}$ in terms of $\mathbf{a}$ and $\mathbf{b}$; [2]
- (b) hence show that the diagonals in a rhombus intersect at right angles. [4]

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7-(MAT/11_HL_Summer_2014_Q12)-Vectors-Lines-Planes

- (a) Show that the points $O(0, 0, 0)$, $A(6, 0, 0)$, $B(6, -\sqrt{24}, \sqrt{12})$, $C(0, -\sqrt{24}, \sqrt{12})$ form a square. [3]
- (b) Find the coordinates of M , the mid-point of $[OB]$. [1]
- (c) Show that an equation of the plane Π , containing the square $OABC$, is $y + \sqrt{2}z = 0$. [3]
- (d) Find a vector equation of the line L , through M , perpendicular to the plane Π . [3]
- (e) Find the coordinates of D , the point of intersection of the line L with the plane whose equation is $y = 0$. [3]
- (f) Find the coordinates of E , the reflection of the point D in the plane Π . [3]
- (g) (i) Find the angle $\hat{ODA}$.
(ii) State what this tells you about the solid $OABCDE$. [6]



8-(MAT/12_HL_Summer_2014_Q12)-Vectors-Lines-Planes

Given the points $A(1, 0, 4)$, $B(2, 3, -1)$ and $C(0, 1, -2)$,

- (a) find the vector equation of the line L_1 passing through the points A and B . [2]

The line L_2 has Cartesian equation $\frac{x-1}{3} = \frac{y+2}{1} = \frac{z-1}{-2}$.

- (b) Show that L_1 and L_2 are skew lines. [5]

Consider the plane Π_1 , parallel to both lines L_1 and L_2 . Point C lies in the plane Π_1 .

- (c) Find the Cartesian equation of the plane Π_1 . [4]

The line L_3 has vector equation $\mathbf{r} = \begin{pmatrix} 3 \\ 0 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} k \\ 1 \\ -1 \end{pmatrix}$.

The plane Π_2 has Cartesian equation $x + y = 12$.

The angle between the line L_3 and the plane Π_2 is 60° .

- (d) (i) Find the value of k .
(ii) Find the point of intersection P of the line L_3 and the plane Π_2 . [7]



9-(MAT/10_HL_Winter_2014_Q3)-Vectors-Lines-Planes

[Maximum mark: 5]

A point P, relative to an origin O, has position vector $\vec{OP} = \begin{pmatrix} 1+s \\ 3+2s \\ 1-s \end{pmatrix}$, $s \in \mathbb{R}$.

Find the minimum length of $\vec{OP}$.

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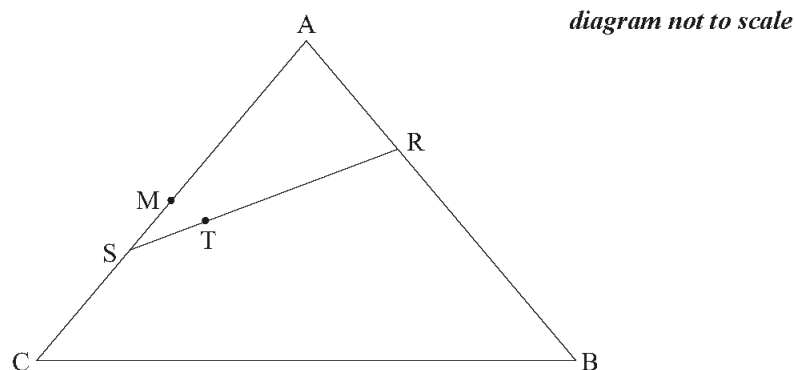
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10-(MAT/10_HL_Winter_2014_Q12)-Vectors-Lines-Planes

The position vectors of the points A, B and C are $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ respectively, relative to an origin O. The following diagram shows the triangle ABC and points M, R, S and T.



M is the midpoint of [AC].

R is a point on [AB] such that $\vec{AR} = \frac{1}{3} \vec{AB}$.

S is a point on [AC] such that $\vec{AS} = \frac{2}{3} \vec{AC}$.

T is a point on [RS] such that $\vec{RT} = \frac{2}{3} \vec{RS}$.

(a) (i) Express $\vec{AM}$ in terms of $\mathbf{a}$ and $\mathbf{c}$.

(ii) Hence show that $\vec{BM} = \frac{1}{2} \mathbf{a} - \mathbf{b} + \frac{1}{2} \mathbf{c}$. [4]

(b) (i) Express $\vec{RA}$ in terms of $\mathbf{a}$ and $\mathbf{b}$.

(ii) Show that $\vec{RT} = -\frac{2}{9} \mathbf{a} - \frac{2}{9} \mathbf{b} + \frac{4}{9} \mathbf{c}$. [5]

(c) Prove that T lies on [BM]. [5]



11-(MAT/11_HL_Summer_2015_Q13)-Vectors-Lines-Planes

Two lines l_1 and l_2 are given respectively by the equations $\vec{r}_1 = \vec{OA} + \lambda \vec{v}$ and $\vec{r}_2 = \vec{OB} + \mu \vec{w}$ where $\vec{OA} = \vec{i} + 2\vec{j} + 3\vec{k}$, $\vec{v} = \vec{i} + \vec{j} + \vec{k}$, $\vec{OB} = 2\vec{i} + \vec{j} - \vec{k}$, $\vec{w} = \vec{i} - \vec{j} + 2\vec{k}$ and O is the origin. Let P be a point on l_1 and let Q be a point on l_2 .

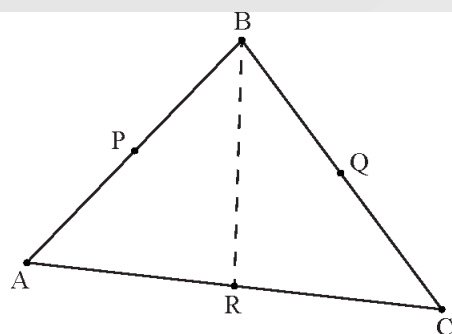
- (a) Find $\vec{PQ}$, in terms of λ and μ . [2]
- (b) Find the value of λ and the value of μ for which $\vec{PQ}$ is perpendicular to the direction vectors of both l_1 and l_2 . [5]
- (c) Hence find the shortest distance between l_1 and l_2 . [3]
- (d) Find the Cartesian equation of the plane Π , which contains line l_1 and is parallel to the direction vector of line l_2 . [5]

Let $\vec{OT} = \vec{OB} + \eta (\vec{v} \times \vec{w})$.

- (e) Find the value of η for which the point T lies in the plane Π . [2]
- (f) For this value of η , calculate $\left| \vec{BT} \right|$. [2]
- (g) State what you notice about your answers to (c) and (f), and give a geometrical interpretation of this result. [2]



12-(MAT/10_HL_Winter_2015_Q13)-Vectors-Lines-Planes



Consider the triangle ABC . The points P , Q and R are the midpoints of the line segments $[AB]$, $[BC]$ and $[AC]$ respectively.

Let $\vec{OA} = \mathbf{a}$, $\vec{OB} = \mathbf{b}$ and $\vec{OC} = \mathbf{c}$.

(a) Find $\vec{BR}$ in terms of $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$. [2]

(b) (i) Find a vector equation of the line that passes through B and R in terms of $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ and a parameter λ .

(ii) Find a vector equation of the line that passes through A and Q in terms of $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ and a parameter μ .

(iii) Hence show that $\vec{OG} = \frac{1}{3}(\mathbf{a} + \mathbf{b} + \mathbf{c})$ given that G is the point where $[BR]$ and $[AQ]$ intersect. [9]

(c) Show that the line segment $[CP]$ also includes the point G . [3]

The coordinates of the points A , B and C are $(1, 3, 1)$, $(3, 7, -5)$ and $(2, 2, 1)$ respectively.

A point X is such that $[GX]$ is perpendicular to the plane ABC .

(d) Given that the tetrahedron $ABCX$ has volume 12 units^3 , find possible coordinates of X . [9]



13-(MAT/11_HL_Summer_2016_Q8)-Vectors-Lines-Planes

[Maximum mark: 5]

O, A, B and C are distinct points such that $\vec{OA} = \mathbf{a}$, $\vec{OB} = \mathbf{b}$ and $\vec{OC} = \mathbf{c}$.

It is given that $\mathbf{c}$ is perpendicular to $\vec{AB}$ and $\mathbf{b}$ is perpendicular to $\vec{AC}$.

Prove that $\mathbf{a}$ is perpendicular to $\vec{BC}$.

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14-(MAT/12_HL_Summer_2016_Q10)-Vectors-Lines-Planes

A line L has equation $\frac{x-2}{p} = \frac{y-q}{2} = z-1$ where $p, q \in \mathbb{R}$.

A plane Π has equation $x + y + 3z = 9$.

(a) Show that L is not perpendicular to Π . [3]

(b) Given that L lies in the plane Π , find the value of p and the value of q . [4]

Consider the different case where the acute angle between L and Π is θ

where $\theta = \arcsin\left(\frac{1}{\sqrt{11}}\right)$.

(c) (i) Show that $p = -2$.

(ii) If L intersects Π at $z = -1$, find the value of q . [11]



15-(MAT/11_HL_Summer_2016_Q11)-Vectors-Lines-Planes

Two planes have equations

$$\Pi_1: 4x + y + z = 8 \text{ and } \Pi_2: 4x + 3y - z = 0$$

- (a) Find the cosine of the angle between the two planes in the form $\sqrt{\frac{p}{q}}$ where $p, q \in \mathbb{Z}$. [4]

Let L be the line of intersection of the two planes.

- (b) (i) Show that L has direction $\begin{pmatrix} -1 \\ 2 \\ 2 \end{pmatrix}$.
- (ii) Show that the point $A(1, 0, 4)$ lies on both planes.
- (iii) Write down a vector equation of L . [6]

B is the point on Π_1 with coordinates $(a, b, 1)$.

- (c) Given the vector $\vec{AB}$ is perpendicular to L find the value of a and the value of b . [5]
- (d) Show that $AB = 3\sqrt{2}$. [1]

The point P lies on L and $\hat{ABP} = 45^\circ$.

- (e) Find the coordinates of the two possible positions of P . [5]

